Complete DSA Patterns Reference Guide

50 Essential Algorithmic Patterns for Coding Interviews & Competitive Programming

Updated: September 2025 | **Problems:** 120+ LeetCode Links

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🎯 Core Problem-Solving Patterns (1-31)

No.	Pattern	Description	Key Idea	Example Problems & Links
1	**Sliding Window**	Used for problems involving subarrays or substrings.	*Use a sliding window to optimize time complexity from $O(n^2)$ to $O(n)$.*	3. Longest Substring Without Repeating Characters 209. Minimum Size Subarray Sum
2	**Two Pointers**	Used for problems involving sorted arrays, linked lists, or string manipulation.	*Use two pointers moving towards/away from each other.*	167. Two Sum II - Input Array Is Sorted 42. Trapping Rain Water
3	**Fast and Slow Pointers**	Used for cyclic problems like finding loops in linked lists.	*Use two pointers moving at different speeds.*	141. Linked List Cycle 876. Middle of the Linked List
4	**Merge Intervals**	Used for problems involving overlapping intervals.	*Sort intervals and merge based on conditions.*	56. Merge Intervals 57. Insert Interval
5	**Cyclic Sort**	Used for problems involving numbers in a given range (1 to n).	*Place each number at its correct index.*	268. Missing Number 442. Find All Duplicates in an Array
6	**Subsets**	Used for problems requiring all combinations/sub sets of elements.	*Use BFS, recursion, or bitmasking.*	78. Subsets 46. Permutations
7	**Binary Search**	Used for searching in sorted arrays or answer-based problems.	*Divide and conquer; halve the search space.*	33. Search in Rotated Sorted Array 704. Binary Search
8	**Backtracking**	Used for constraint-based problems like permutations and combinations.	*Try all possibilities and backtrack upon failure.*	51. N-Queens 79. Word Search
9	**Breadth-First Search (BFS)**	Used for shortest path or level-order traversal problems.	*Explore all neighbors at the current level before moving to the next level.*	102. Binary Tree Level Order Traversal 127. Word Ladder
10	**Depth-First Search (DFS)**	Used for pathfinding, tree/graph traversal and connected components.	*Recursively or iteratively explore each path fully.*	797. All Paths From Source to Target 200. Number of Islands

No.	Pattern	Description	Key Idea	Example Problems & Links
11	**Topological Sort**	Used for problems with dependencies in a Directed Acyclic Graph (DAG).	*Use BFS or DFS to order tasks based on prerequisites.*	207. Course Schedule 210. Course Schedule II
12	**Union-Find (Disjoint Set)**	Used for connectivity in graphs.	*Use union and find operations to manage connected components.*	547. Number of Provinces 684. Redundant Connection
13	**Greedy**	Used for optimization problems (minimizing/maximizing something).	*Make the locally optimal choice at each step.*	435. Non-overlapping Intervals 621. Task Scheduler
14	**Dynamic Programming (DP)**	Used for optimization and decision-based problems.	*Break the problem into overlapping subproblems.*	300. Longest Increasing Subsequence 416. Partition Equal Subset Sum
15	**Bit Manipulation**	Used for binary-related problems like subsets and XOR queries.	*Use bitwise operators to solve efficiently.*	136. Single Number 231. Power of Two
16	**Matrix Traversal**	Used for problems involving grid traversal.	*Use BFS, DFS, or dynamic programming.*	62. Unique Paths 994. Rotting Oranges
17	**Heap / Priority Queue**	Used for problems requiring frequent max/min operations.	*Use heaps for efficient insertion and extraction.*	215. Kth Largest Element in an Array 23. Merge k Sorted Lists
18	**Divide and Conquer**	Used for problems involving partitioning.	*Divide the problem into smaller subproblems.*	912. Sort an Array 4. Median of Two Sorted Arrays
19	**Prefix Sum**	Used for problems involving range sums.	*Precompute cumulative sums to optimize queries.*	560. Subarray Sum Equals K 303. Range Sum Query - Immutable
20	**Sliding Window Maximum**	Used for problems involving maximum/minimum in sliding windows.	*Use deque to maintain window maximum efficiently.*	239. Sliding Window Maximum 1425. Constrained Subsequence Sum

No.	Pattern	Description	Key Idea	Example Problems & Links
21	**Kadane's Algorithm**	Used for maximum subarray problems.	*Maintain a running sum and update the max sum.*	53. Maximum Subarray 918. Maximum Sum Circular Subarray
22	**Trie (Prefix Tree)**	Used for word-related problems.	*Use a tree structure for fast prefix lookups.*	208. Implement Trie (Prefix Tree) 212. Word Search II
23	**Segment Trees**	Used for range query problems.	*Build a tree structure for efficient range queries.*	307. Range Sum Query - Mutable 315. Count of Smaller Numbers After Self
24	**Graph Traversal**	Used for graph-related problems like shortest paths or connected components.	*Use DFS, BFS, or Dijkstra's algorithm.*	743. Network Delay Time 1584. Min Cost to Connect All Points
25	**Flood Fill**	Used for grid-based coloring or connected regions.	*Use DFS or BFS to visit all connected components.*	733. Flood Fill 1020. Number of Enclaves
26	**Monotonic Stack**	Used for problems involving nearest larger/smaller elements.	*Use a stack to maintain a monotonic sequence.*	496. Next Greater Element I 84. Largest Rectangle in Histogram
27	**String Matching (KMP, Rabin-Karp)**	Used for finding substrings in a string.	*Use efficient string matching algorithms.*	28. Find the Index of the First Occurrence in a String 214. Shortest Palindrome
28	**Binary Indexed Tree (Fenwick Tree)**	Used for dynamic range sum/updates.	*Use a tree structure to efficiently compute prefix sums.*	307. Range Sum Query - Mutable 315. Count of Smaller Numbers After Self
29	**Reservoir Sampling**	Used for random sampling.	*Keep track of k items randomly selected from a stream.*	382. Linked List Random Node 398. Random Pick Index
30	**LRU Cache**	Used for caching problems.	*Use a hashmap and doubly linked list.*	146. LRU Cache 460. LFU Cache

No.	Pattern	Description	Key Idea	Example Problems & Links
31	**Fibonacci Sequence**	Used for DP problems.	*Compute Fibonacci numbers iteratively or using matrix exponentiation.*	70. Climbing Stairs 198. House Robber

⚡ Advanced Patterns (32-50)

No.	Pattern	Description	Key Idea	Example Problems & Links
32	**Morris Traversal**	Used for tree traversal without extra space.	*Use threading to traverse without recursion/stack.*	94. Binary Tree Inorder Traversal 144. Binary Tree Preorder Traversal
33	**Boyer-Moore Majority Vote**	Used for finding majority elements.	*Use voting algorithm to find majority in O(n) time.*	169. Majority Element 229. Majority Element II
34	**Rolling Hash**	Used for string/array comparison problems.	*Use polynomial rolling hash for efficient comparisons.*	187. Repeated DNA Sequences 1044. Longest Duplicate Substring
35	**Manacher's Algorithm**	Used for palindrome problems.	*Find all palindromes in linear time.*	5. Longest Palindromic Substring 647. Palindromic Substrings
36	**Catalan Numbers**	Used for counting problems with nested structures.	*Count valid combinations of nested elements.*	22. Generate Parentheses 96. Unique Binary Search Trees
37	**Game Theory (Minimax)**	Used for game-playing problems.	*Use minimax with alpha-beta pruning.*	464. Can I Win 877. Stone Game
38	**Line Sweep**	Used for computational geometry problems.	*Process events in sorted order.*	253. Meeting Rooms II 218. The Skyline Problem
39	**Shortest Path Algorithms**	Used for graph path problems.	*Dijkstra's, Bellman-Ford, Floyd-Warshall.*	787. Cheapest Flights Within K Stops 1334. Find the City With Smallest Number of Neighbors
40	**Meet in the Middle**	Used for optimization when brute force is O(2^n).	*Split problem space and combine results.*	18. 4Sum 1755. Closest Subsequence Sum
41	**Critical Connections**	Used for finding critical nodes/edges in graphs.	*Use DFS with low-link values (Tarjan's Algorithm).*	1192. Critical Connections in a Network 1568. Minimum Days to Disconnect Island

No.	Pattern	Description	Key Idea	Example Problems & Links
42	**Z-Algorithm**	Used for string matching problems.	*Find all occurrences of pattern in text in linear time.*	28. Find the Index of the First Occurrence 1392. Longest Happy Prefix
43	**Coordinate Compression**	Used when dealing with large coordinate ranges.	*Map large coordinates to smaller range.*	391. Perfect Rectangle 850. Rectangle Area II
44	**Convex Hull**	Used for computational geometry problems.	*Find the convex hull of a set of points.*	587. Erect the Fence 1453. Maximum Darts Inside Circular Dartboard
45	**Sqrt Decomposition**	Used for range queries with updates.	*Divide array into $\sqrt{n}$ blocks for balanced operations.*	307. Range Sum Query - Mutable 327. Count of Range Sum
46	**Heavy-Light Decomposition**	Used for tree path queries (Advanced).	*Decompose tree into heavy and light edges.*	Used in competitive programming Tree path sum queries
47	**Network Flow (Max Flow)**	Used for flow optimization problems.	*Find maximum flow through a network.*	Maximum bipartite matching Min-cut problems
48	**Persistent Data Structures**	Used when you need to maintain multiple versions.	*Keep history of data structure modifications.*	Version control systems Functional programming structures
49	**Suffix Array/Tree**	Used for string processing problems.	*Efficient substring operations and pattern matching.*	1044. Longest Duplicate Substring Advanced string algorithms
50	**Aho-Corasick Algorithm**	Used for multiple string matching.	*Build automaton for multiple pattern matching.*	1032. Stream of Characters Multiple pattern search

📊 Time and Space Complexity Quick Reference

Algorithm/Data Structure	Time Complexity	Space Complexity	Use Cases
Binary Search	O(log n)	O(1)	Sorted arrays, answer-based problems
Merge Sort	O(n log n)	O(n)	Stable sorting, external sorting
Quick Sort	O(n log n) avg, $O(n^2)$ worst	O(log n)	In-place sorting, average case performance
Heap Operations	O(log n) insert/delete, O(1) peek	O(n) for heap	Priority queues, top-k problems
Hash Table	O(1) avg, O(n) worst	O(n)	Fast lookups, counting, caching
DFS/BFS	O(V + E)	O(V)	Graph traversal, tree operations
Dijkstra's Algorithm	O((V + E) log V)	O(V)	Single source shortest path
Union-Find	O(α(n)) amortized	O(n)	Dynamic connectivity, MST
Trie Operations	O(m) where m is key length	O(ALPHABET_SIZE × N × M)	Prefix matching, autocomplete
Segment Tree	O(log n) query/update	O(n)	Range queries with updates

🎯 Pattern Selection Guide

📚 For Array/String Problems:

- **Sliding Window:** Subarray/substring problems with contiguous elements
- **Two Pointers:** Sorted arrays, palindromes, or meeting from ends
- **Prefix Sum:** Range sum queries, subarray sum problems
- **Binary Search:** Sorted arrays, answer-based problems

🌳 For Tree/Graph Problems:

- **DFS:** Path problems, tree traversal, backtracking
- **BFS:** Shortest path, level-order traversal, minimum steps
- **Union-Find:** Connectivity problems, dynamic components
- **Topological Sort:** Dependency resolution, course scheduling

⚡ For Optimization Problems:

- **Dynamic Programming:** Overlapping subproblems, optimal substructure

- **Greedy:** Locally optimal choices lead to global optimum
- **Backtracking:** Constraint satisfaction, generate all solutions

🏗️ For Data Structure Design:

- **Heap:** Priority-based operations, top-k problems
- **Trie:** Word/prefix problems, autocomplete
- **Segment Tree:** Range queries with updates
- **Hash Map:** Fast lookups, counting, caching

📝 Study Plan Recommendations

🚀 Beginner Level (Patterns 1-15):

Start with fundamental patterns that appear in 70% of coding interviews. Master these before moving to advanced topics.

🎯 Intermediate Level (Patterns 16-31):

Focus on data structure design and graph algorithms. These patterns are common in system design and complex algorithmic problems.

🏆 Advanced Level (Patterns 32-50):

Specialized algorithms for competitive programming and senior-level interviews. Master these for challenging technical roles.

📈 Practice Strategy:

- Solve 3-5 problems per pattern before moving to the next
- Focus on pattern recognition rather than memorizing solutions
- Time yourself: aim for 20-30 minutes per medium problem
- Review and optimize your solutions
- Practice explaining your approach out loud